# Chapter 9 — Electrostatics and Current Electricity

Board-style MCQs · PECTAA Physics 11 · Answer key at end

## Practice MCQs

**1. Coulomb’s law is:**

**(a)** F = qr/k **(b)** F = $kq_1q_2r^2$ **(c)** F = $kq_1/q_2r$ **(d)** F = $kq_1q_2/r^2$

**2. The value of k in free space is about:**

**(a)** $9 \times 10^{-9}$ N m² C⁻² **(b)** $6 \times 10^{-11}$ N m² C⁻² **(c)** $1.6 \times 10^{-19}$ N m² C⁻² **(d)** $9 \times 10^{9}$ N m² C⁻²

**3. Like charges:**

**(a)** do not interact **(b)** repel **(c)** attract **(d)** neutralize

**4. Electric field strength is:**

**(a)** F + q **(b)** q/F **(c)** Fq **(d)** F/q

**5. SI unit of electric field is:**

**(a)** J C⁻¹ **(b)** V m **(c)** C N⁻¹ **(d)** N C⁻¹

**6. Electric field due to point charge is:**

**(a)** kq/r **(b)** $kqr^2$ **(c)** $kq/r^2$ **(d)** $q/kr^2$

**7. Electric field between parallel plates is:**

**(a)** V + d **(b)** d/V **(c)** V/d **(d)** Vd

**8. Electric flux is:**

**(a)** EA cos θ **(b)** E/A **(c)** A/E **(d)** EA sin θ

**9. Gauss’s law gives total flux as:**

**(a)** $\varepsilon_0/Q$ **(b)** $Q\varepsilon_0$ **(c)** $Q + \varepsilon_0$ **(d)** $Q/\varepsilon_0$

**10. Electric field inside a charged conductor is:**

**(a)** $kq/r^2$ **(b)** zero **(c)** V/d **(d)** maximum

**11. Outside a charged sphere, field is as if charge is at:**

**(a)** infinity **(b)** centre **(c)** edge **(d)** surface

**12. Electric potential is:**

**(a)** W + q **(b)** q/W **(c)** Wq **(d)** W/q

**13. SI unit of potential is:**

**(a)** volt **(b)** ohm **(c)** ampere **(d)** coulomb

**14. Relation between E and potential gradient is:**

**(a)** E = d/V **(b)** E = Vd **(c)** E = $\Delta V/\Delta d$ **(d)** E = $-\Delta V/\Delta d$

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**15. One electron volt equals:**

**(a)** $1.6 \times 10^{-19}$ J **(b)** 1 V **(c)** $1.6 \times 10^{-19}$ C **(d)** 1 C

**16. Work done on electron through 100 V is:**

**(a)** 1 eV **(b)** $1.6 \times 10^{-19}$ eV **(c)** 100 eV **(d)** 100 J

**17. A charged particle entering uniform E perpendicularly follows:**

**(a)** ellipse **(b)** parabola **(c)** circle **(d)** straight line always

**18. A Faraday cage shields because:**

**(a)** it has no mass **(b)** it absorbs light **(c)** induced charges cancel internal field **(d)** it is magnetic

**19. Electric current is:**

**(a)** Q + t **(b)** Q/t **(c)** Qt **(d)** t/Q

**20. Drift velocity relation is:**

**(a)** I = nAvq **(b)** I = nA/qv **(c)** I = Av/nq **(d)** I = q/nAv

**21. Ohm's law is:**

**(a)** V = I/R **(b)** V = I + R **(c)** V = R/I **(d)** V = IR

**22. Resistance of a wire is:**

**(a)** ρA/L **(b)** LA/ρ **(c)** ρ/LA **(d)** ρL/A

**23. Resistivity depends on:**

**(a)** shape only **(b)** length only **(c)** area only **(d)** material and temperature

**24. In metals, resistance with temperature:**

**(a)** decreases **(b)** remains same **(c)** increases **(d)** becomes zero

**25. Electrical power is:**

**(a)** V/I **(b)** I/V **(c)** VI **(d)** V + I

**26. Power can also be written as:**

**(a)** V/R **(b)** R/I **(c)** I/R **(d)** $I^2R$

**27. EMF is equal to terminal voltage when:**

**(a)** resistance is zero **(b)** temperature is zero **(c)** current is maximum **(d)** no current flows

**28. Terminal voltage is:**

**(a)** E/Ir **(b)** E − Ir **(c)** E + Ir **(d)** Ir − E

**29. Kirchhoff's first rule is conservation of:**

**(a)** mass **(b)** momentum **(c)** energy **(d)** charge

**30. Kirchhoff's second rule is conservation of:**

**(a)** flux **(b)** energy **(c)** momentum **(d)** charge

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**31. Wheatstone bridge balances when:**

**(a)** $R_1/R = R_2/R_4$ **(b)** $R_1 + R_2 = R_3 + R$ **(c)** $R_1R_2 = R_3R_4$ **(d)** $R_1/R_2 = R_3/R_4$

**32. At balanced Wheatstone bridge, galvanometer current is:**

**(a)** Ir **(b)** maximum **(c)** zero **(d)** E/R

**33. Potentiometer works as:**

**(a)** current source **(b)** magnetic sensor **(c)** Wheatstone bridge only **(d)** potential divider with null method

**34. Potentiometer balance gives:**

**(a)** $E_1/E_2 = \ell_1/\ell_2$ **(b)** $E_1/E_2 = \ell_2/\ell_1$ **(c)** $E_1\ell_1 = E_2\ell_2$ **(d)** $E_1 + E_2 = \ell_1 + \ell_2$

**35. Most thermistors when heated:**

**(a)** lose charge **(b)** lose resistance **(c)** gain resistance **(d)** become superconductors

**36. An LDR when illuminated:**

**(a)** becomes insulator **(b)** loses resistance **(c)** loses charge **(d)** gains resistance

**37. A bulb lights instantly because:**

**(a)** electrons move at light speed **(b)** field is established quickly **(c)** current is stored **(d)** protons flow

**38. Conventional current direction is:**

**(a)** same as electron flow **(b)** positive to negative outside source **(c)** random **(d)** negative to positive outside source

**39. A battery rated 100 A h supplies charge:**

**(a)** $3.6 \times 10^5$ C **(b)** $1.6 \times 10^{-19}$ C **(c)** 3600 C **(d)** 100 C

**40. If wire length doubles and area halves, resistance becomes:**

**(a)** R/4 **(b)** R/2 **(c)** 2R **(d)** 4R

**41. If wire is doubled on itself, resistance becomes:**

**(a)** R/2 **(b)** 2R **(c)** R/4 **(d)** 4R

**42. A 500 W, 220 V bulb has filament resistance:**

**(a)** same **(b)** lower than 100 W bulb **(c)** zero **(d)** higher than 100 W bulb

**43. Carbon fibres in concrete help detect:**

**(a)** sound only **(b)** light only **(c)** cracks and damage **(d)** temperature only

**44. Electric field lines never cross because:**

**(a)** they are sound waves **(b)** field has one direction at a point **(c)** they are magnetic **(d)** they have no energy

**45. Net flux through uncharged hollow sphere in external field is:**

**(a)** maximum **(b)** $Q/\varepsilon_0$ **(c)** negative **(d)** zero

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**46. Galaxy is almost electrically neutral because:**

**(a)** magnetic force cancels **(b)** electric force is very strong **(c)** charges do not exist **(d)** gravity is stronger

**47. Increasing potential difference across wire:**

**(a)** increases drift velocity **(b)** decreases drift velocity **(c)** does not change drift **(d)** stops current

**48. Terminal voltage decreases when current increases because:**

**(a)** internal drop Ir increases **(b)** resistance decreases **(c)** charge decreases **(d)** emf increases

**49. A thermistor is used in:**

**(a)** only magnetic sensing **(b)** only light sensing **(c)** only sound sensing **(d)** temperature sensing

**50. An LDR is used in:**

**(a)** pressure sensing only **(b)** magnetic sensing only **(c)** light sensing **(d)** temperature sensing only

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## Answer Key with Brief Reasons

**1. Answer: (d) $F = kq_1q_2/r^2$** — Inverse-square electric force.

**2. Answer: (d) $9 \times 10^9$ N m² C⁻²** — $k = 1/4\pi\varepsilon_0$.

**3. Answer: (b) repel** — Same sign repels.

**4. Answer: (d) F/q** — Force per unit charge.

**5. Answer: (d) N C⁻¹** — E = F/q.

**6. Answer: (c) $kq/r^2$** — Point charge field.

**7. Answer: (c) V/d** — Uniform field E = V/d.

**8. Answer: (a) EA cos θ** — Flux = field × perpendicular area.

**9. Answer: (d) $Q/\varepsilon_0$** — $\Phi = Q_{enclosed}/\varepsilon_0$.

**10. Answer: (b) zero** — Charges reside on surface.

**11. Answer: (b) centre** — Spherical symmetry.

**12. Answer: (d) W/q** — Work per unit charge.

**13. Answer: (a) volt** — 1 V = 1 J/C.

**14. Answer: (d) $E = -\Delta V/\Delta d$** — Field points down potential.

**15. Answer: (a) $1.6 \times 10^{-19}$ J** — Energy of electron through 1 V.

**16. Answer: (c) 100 eV** — Energy = qV = 100 eV.

**17. Answer: (b) parabola** — Constant transverse acceleration.

**18. Answer: (c) induced charges cancel internal field** — Conducting enclosure cancels field.

**19. Answer: (b) Q/t** — Rate of flow of charge.

**20. Answer: (a) I = nAvq** — Current from carrier density.

**21. Answer: (d) V = IR** — Voltage-current relation.

**22. Answer: (d) ρL/A** — $R = \rho L/A$.

**23. Answer: (d) material and temperature** — It is a material property.

**24. Answer: (c) increases** — Lattice scattering increases.

**25. Answer: (c) VI** — P = VI.

**26. Answer: (d) $I^2R$** — $P = I^2R$.

**27. Answer: (d) no current flows** — No internal drop.

**28. Answer: (b) E − Ir** — Internal resistance drops voltage.

**29. Answer: (d) charge** — Junction current rule.

**30. Answer: (b) energy** — Loop voltage rule.

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**31. Answer: (d) $R_1/R_2 = R_3/R_4$** — No galvanometer current.

**32. Answer: (c) zero** — Null condition.

**33. Answer: (d) potential divider with null method** — It compares emfs by balancing length.

**34. Answer: (a) $E_1/E_2 = \ell_1/\ell_2$** — Emf proportional to length.

**35. Answer: (b) lose resistance** — NTC thermistors decrease R.

**36. Answer: (b) loses resistance** — Light increases carriers.

**37. Answer: (b) field is established quickly** — All electrons drift together.

**38. Answer: (b) positive to negative outside source** — Opposite to electron drift.

**39. Answer: (a) $3.6 \times 10^5$ C** — Q = 100 × 3600.

**40. Answer: (d) 4R** — R ∝ L/A.

**41. Answer: (c) R/4** — Length half, area double.

**42. Answer: (b) lower than 100 W bulb** — $R = V^2/P$.

**43. Answer: (c) cracks and damage** — Resistance changes with damage.

**44. Answer: (b) field has one direction at a point** — Two directions impossible.

**45. Answer: (d) zero** — No enclosed charge.

**46. Answer: (b) electric force is very strong** — Large charge imbalance would dominate.

**47. Answer: (a) increases drift velocity** — E and drift increase.

**48. Answer: (a) internal drop Ir increases** — V = E − Ir.

**49. Answer: (d) temperature sensing** — Resistance depends on temperature.

**50. Answer: (c) light sensing** — Resistance depends on light.

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